Jason Xinyu Liu

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Academic Appointments

2025 – present **Massachusetts Institute of Technology**
Postdoctoral Associate in Aeronautics and Astronautics
Advisor: Prof. Julie Shah

Education

2018 – 2025 **Brown University**
Ph.D. in Computer Science
Advisor: Prof. Stefanie Tellex
Dissertation: Robotic Language Grounding [PDF, Video]

2021 **Brown University**
M.S. in Computer Science
Advisor: Prof. Stefanie Tellex

2017 **University of California, Berkeley**
B.S. in Electrical Engineering and Computer Sciences
Advisor: Prof. Ken Goldberg and Prof. David Wagner

Fellowships, Awards and Honors

Brown **National Science Foundation Graduate Research Fellowship Program**, 2018
Jack Kent Cooke Foundation Graduate Scholarship, 2018

UC Berkeley **Tau Beta Pi, National Engineering Honor Society**, 2014
Eta Kappa Nu, National EECS Honor Society, 2014
Jack Kent Cooke Foundation Undergraduate Transfer Scholarship, 2014

Publications

CoRL 2026 **Language as a Sensor: Calibrated Spatial Belief Estimation in 3D Scenes from Natural Language**
A. Naveen, **J.X. Liu**, L. Carlone, A. Bobu
Conference on Robot Learning (CoRL) 2026. [Paper]

RA-L 2026 **GHOST in the Robots: Real-Time Exocentric Dual-Robot VR Teleoperation from Onboard Cameras**
Y. Wei, F. Zaghloul, S.C. Aryal, A.K. Agrawal, C. Li, **J.X. Liu**, J. Tompkin, S. Tellex
IEEE Robotics and Automation Letters (RA-L) 2026. [Paper]

ICRA 2026 **Pragmatic Embodied Spoken Instruction Following in Human-Robot Collaboration with Theory of Mind**
L. Ying, X. Li, S. Aarya, Y. Fang, Y. Yin, **J.X. Liu**, S. Tellex, J.B. Tenenbaum, T. Shu
International Conference on Robotics and Automation (ICRA) 2026. [Paper]

HRI 2026 **LEGS-POMDP: Language and Gesture-Guided Object Search in Partially Observable Environments**
I. He, S. Tellex, **J.X. Liu**
ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026. [Paper, Website]

RA-L 2025 **Open-vocabulary Pick and Place via Patch-level Semantic Maps**
M.Jia, H. Huang, Z. Zhang, C. Wang, L. Zhao, D. Wang, **J.X. Liu**, R. Walters , R. Platt , S. Tellex
IEEE Robotics and Automation Letters (RA-L) 2025. Best Paper Award Honorable Mention. [Paper, Website]

IROS 2024 **Lang2LTL-2: Grounding Spatiotemporal Navigation Commands Using Large Language and Vision-Language Models**
J.X. Liu, A. Shah, G. Konidaris, S. Tellex, and D. Paulius
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024. [Paper, Website]

IJCAI 2024 **A Survey of Robotic Language Grounding: Tradeoffs between Symbols and Embeddings**
V. Cohen*, **J.X. Liu***, R. Mooney*, S. Tellex*, D. Watkins*
International Joint Conference on Artificial Intelligence (IJCAI) 2024 Survey Track. [Paper]

ICRA 2024 **LTL-Transfer: Skill Transfer for Temporally-Extended Task Specifications**
J.X. Liu*, A. Shah*, E. Rosen, M. Jia, G.D. Konidaris, S. Tellex
IEEE International Conference on Robotics and Automation (ICRA) 2024. [Paper, Website]

CoRL 2023 **Lang2LTL: Grounding Complex Natural Language Commands for Temporal Tasks in Unseen Environments**
J.X. Liu*, Z. Yang*, I. Idrees, S. Liang, B. Schornstein, S. Tellex, A. Shah
Conference on Robot Learning (CoRL) 2023. [Paper, Website, Brown News]
AAAI Fall Symposium on Unifying Representations for Robot Application Development (URRAD) 2023. Best Paper Finalist.

INTERSPEECH 2023 **Style-transfer based Speech and Audio-visual Scene Understanding for Robot Action Sequence Acquisition from Videos**
C. Hori, P. Peng, D. Harwath, **J.X. Liu**, K. Ota, S. Jain, R. Corcodel, D. Jha, D. Romeres, J. Le Roux
INTERSPEECH 2023. [Paper]

ICRA 2022 **Generalizing to New Domains by Mapping Natural Language to Lifted LTL**
E. Hsiung, H. Mehta, J. Chu, **J.X. Liu**, R. Patel, S. Tellex, G. D. Konidaris
IEEE International Conference on Robotics and Automation (ICRA) 2022. [Paper]

ACNS 2020 **Minority Reports Defense: Defending Against Adversarial Patches**
M. McCoyd, W. Park, S. Chen, N. Shah, R. Roggenkemper, M. Hwang, **J.X. Liu**, D. Wagner
Applied Cryptography and Network Security Workshops (ACNS) 2020. [Paper]

ICRA 2018 **Dex-Net 3.0: Computing Robust Vacuum Suction Grasp Targets in Point Clouds Using a New Analytic Model and Deep Learning**
J. Mahler, M. Matl, **J.X. Liu**, A. Li, D. Gealy, K. Goldberg
IEEE International Conference on Robotics and Automation (ICRA) 2018. [Paper, Website]

ICISS 2018 **Detecting Phone Theft Using Machine Learning**
J.X. Liu, D. Wagner, S. Egelman
International Conference on Information Science and System (ICISS) 2018. [Paper]

RSS 2017 **Dex-Net 2.0: Deep Learning to Plan Robust Grasps with Synthetic Point Clouds and Analytic Grasp Metrics**
J. Mahler, J. Liang, S. Niyaz, M. Laskey, R. Doan, **J.X. Liu**, J. A. Ojea, K. Goldberg
Robotics: Science and Systems (RSS) 2017. [Paper, Website]

Invited Talks

October 2026	CoRL 2-26 Workshop Spatial Intelligence and Structured Reasoning for Embodied Navigation (hosted by Dr. Bhatt, Neel)
March 2026	The University of Edinburgh, RL Seminar (hosted by Dr. David Abel)
Oct 2025	UIUC, CS Colloquium [web] (hosted by Prof. Dilek Hakkani-Tür and Prof. Gokhan Tür)
Oct 2025	Tufts, HRI Seminar (hosted by Prof. Jivko Sinapov)
Jan 2025	USC, Robotics Seminar (hosted by Prof. Jesse Thomason)
Dec 2024	Brown, Brown Database Group (hosted by Prof. Ugur Çetintemel)
Oct 2024	Amazon Robotics, Amazon Robotics Research Fall Symposium and PhD Communication Competition [web]
Oct 2024	UT Austin, Forum for Artificial Intelligence [video] (hosted by Prof. Joydeep Biswas)
Oct 2024	CU Boulder, CS Colloquium [video] (hosted by Prof. Bradley Hayes)
Sep 2024	MIT, Interactive Robotics Group (hosted by Prof. Julie Shah)
Sep 2024	UT Austin, Center of Autonomy (hosted by Prof. Ufuk Topcu)
Aug 2024	Arizona State, Logos Robotics Lab (hosted by Prof. Nakul Gopalan)
Jul 2024	Oxford, CS Department Seminar (hosted by Prof. Giuseppe De Giacomo)
Oct 2022	Northeast Robotics Colloquium

Guest Lectures

Spring 2026	Sapienza University of Rome, Hybrid Neurosymbolic Techniques for Non-Markovian Reinforcement Learning
Spring 2025	Brown, CSCI 1470: Deep Learning
Fall 2024	Brown, CSCI 1951X: Formal Proof and Verification

Mentorship

2026	Aryan Naveen (MIT Ph.D.)
2026	Yiran Tao (MIT Ph.D.)
2026 Summer	Amanda Cheng (Wellesley Undergraduate)
2026 Summer	Jonas Xiangzhou Sun (MIT Undergraduate)
2026 Summer	Jefferson Lin (MIT Undergraduate)
2024 - 2025	Ivy He (Brown Ph.D.)
2023 - 2025	Yichen Wei (Brown Master's. Now Ph.D. at UPenn)
2023 - 2024	Anneke Wernerfelt (Haverford Undergraduate. Master's at UPenn)
2021 - 2022	Eric Hsiung (Brown Master's. Now Ph.D. at UT Austin)
Spring 2021	Shihab Khalfalla (UConn Undergraduate. Now at Microsoft)
2020 - 2021	Suchen Zheng (Brown Master's. Now Ph.D. at Yale)
2020 - 2021	Tyler Edwards (UPenn Undergraduate. Master's at UPenn)

Workshop Organization

CoRL 2026	Modeling Uncertainty in Robotic World Models. [Website]
IROS 2026	Leveraging Human-Robot Dialogue to Improve Robot Learning and Human-Robot Interaction. [Website]
IROS 2026	Learning and Formal Methods for Robotics. [Website]

CoRL 2025 Evaluation and Deployment Across the Robot Learning Lifecycle. [Website]

RSS 2025 Robot Planning in the Era of Foundation Models. [Website]

RSS 2024 Task Specification for General-Purpose Intelligent Robots. [Website]

Service

Brown **First-year Ph.D. student mentor**, 2021-2023
Google exploreCSR (explorecsr.cs.brown.edu), Spring 2021

Review **THRI** 2026, **RSS** 2026, **JMLR** 2025, **CoRL-W** 2024, **RSS-W** 2024, **IROS** 2024, **ICRA** 2023, **ICRA** 2022

Industry Experience

Summer 2022 **Research Intern**, *Mitsubishi Electric Research Laboratories*, Cambridge, MA

Summer 2015 **Software Engineer**, *NetSuite*, San Mateo, CA